Wireshark Practical Manual

Practical No. 1: Installation and Interface Overview

Objective

To install Wireshark and understand its graphical user interface.

Requirements

- Windows/Linux/macOS
- Wireshark
- Internet connection

Procedure

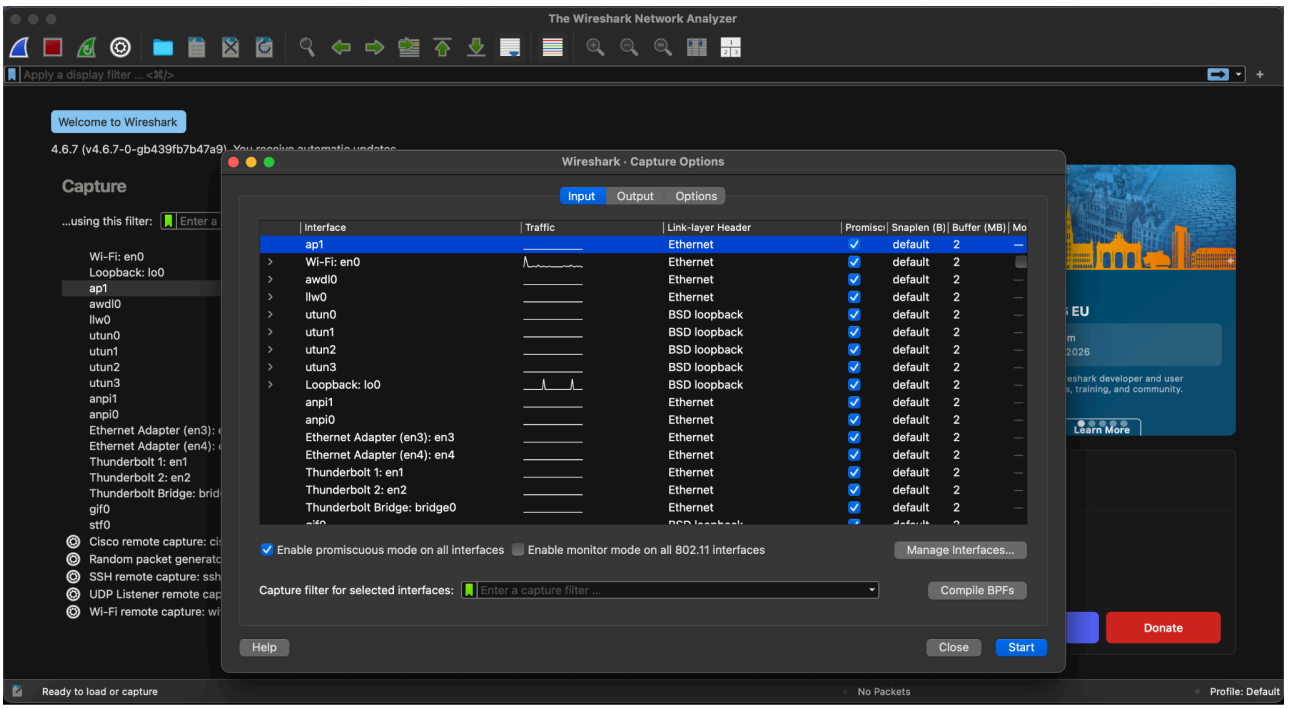
1. Download Wireshark from the official website.
2. Install the software.
3. Launch Wireshark.
4. Observe the available network interfaces.
5. Identify the toolbar, packet list, packet details, and packet bytes panes.

Observation

1. Wireshark was installed and launched successfully.
2. The Capture Options window displays all available network interfaces (Wi-Fi, Loopback, Thunderbolt, etc.).
3. The Wi-Fi (en0) interface shows active network traffic and can be selected for packet capture.

Result

Successfully installed Wireshark and understood its interface.



Practical No. 2: Capturing Network Packets

Objective

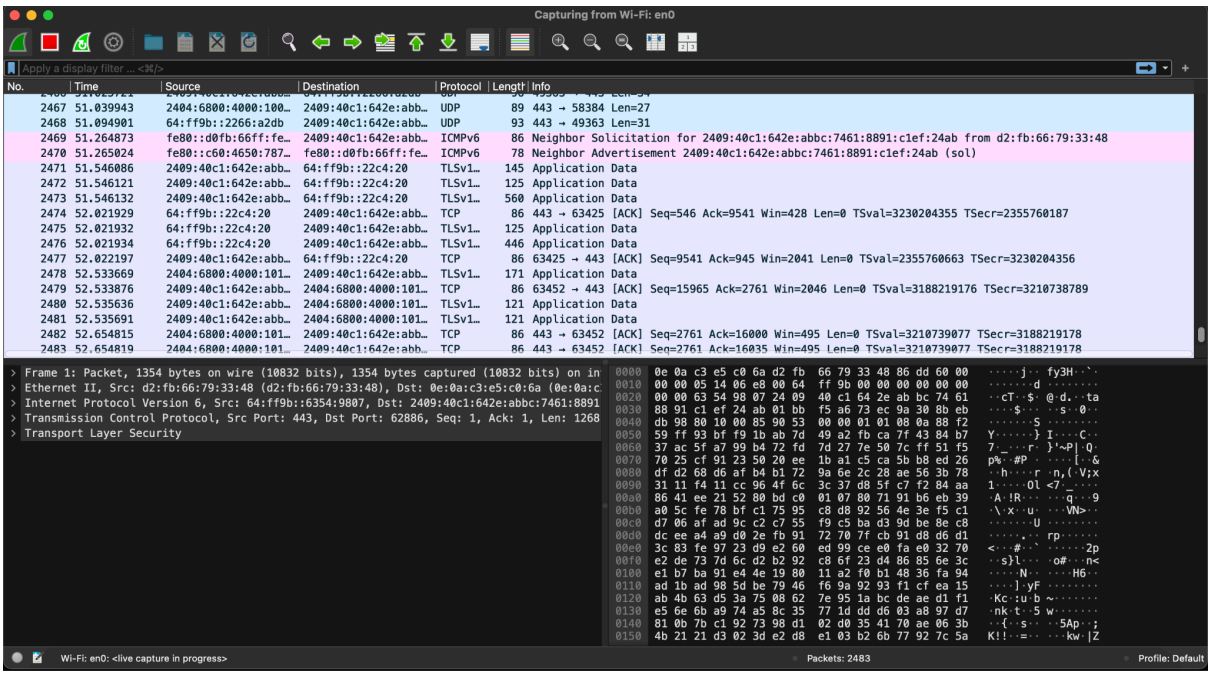
To capture live network packets.

Procedure

1. Open Wireshark.
2. Select the active network interface.
3. Click **Start Capture**.
4. Browse any website.
5. Stop packet capture.
6. Save the capture as **capture.pcapng**.

Result

Successfully captured live network traffic.



Practical No. 3: TCP Three-Way Handshake Analysis

Objective

To analyze TCP connection establishment.

Display Filter

```
tcp
```

Procedure

1. Start packet capture.
2. Open any website.
3. Stop the capture.
4. Apply filter:

```
tcp.flags.syn==1
```

5. Observe:

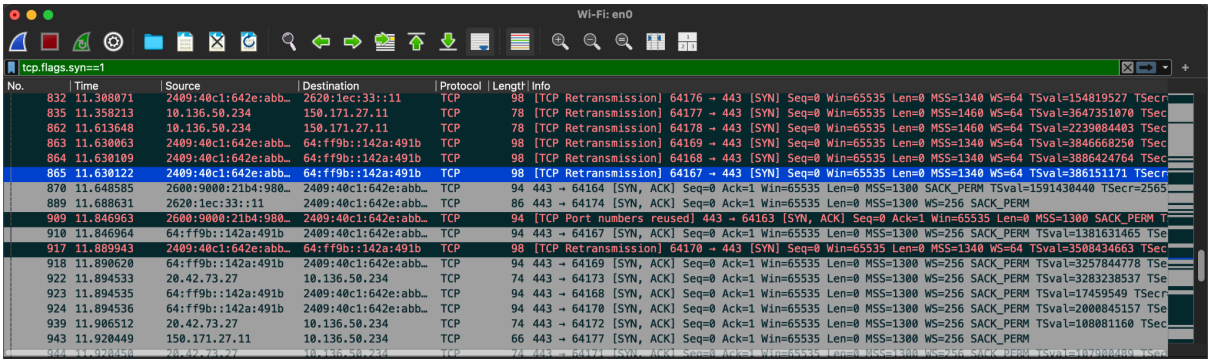
- SYN
- SYN-ACK
- ACK

Observation

TCP establishes communication using a three-way handshake.

Result

Successfully analyzed TCP handshake.



Practical No. 4: DNS Packet Analysis

Objective

To analyze DNS queries.

Display Filter

```
dns
```

Procedure

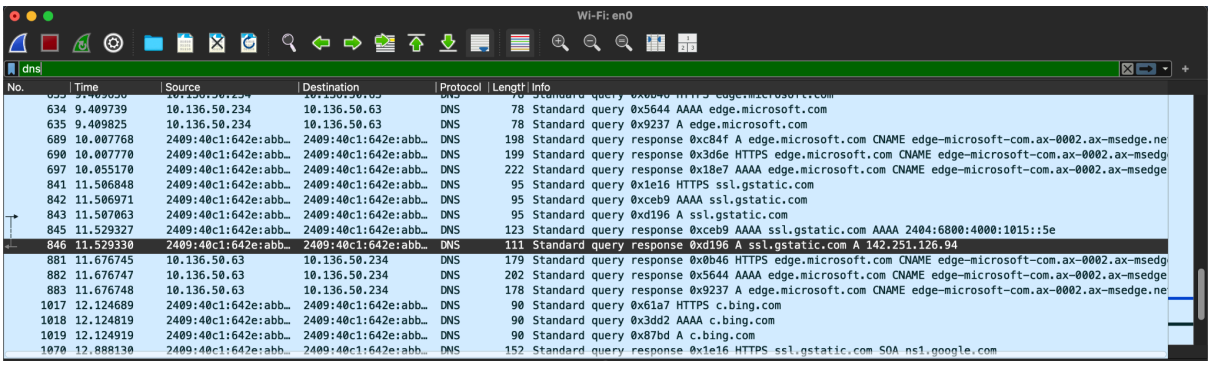
1. Start capturing.
2. Visit a new website.
3. Stop capture.
4. Apply DNS filter.
5. Observe query and response packets.

Observation

DNS converts domain names into IP addresses.

Result

Successfully analyzed DNS packets.



Practical No. 5: HTTP Traffic Analysis

Objective

To analyze HTTP protocol.

Display Filter

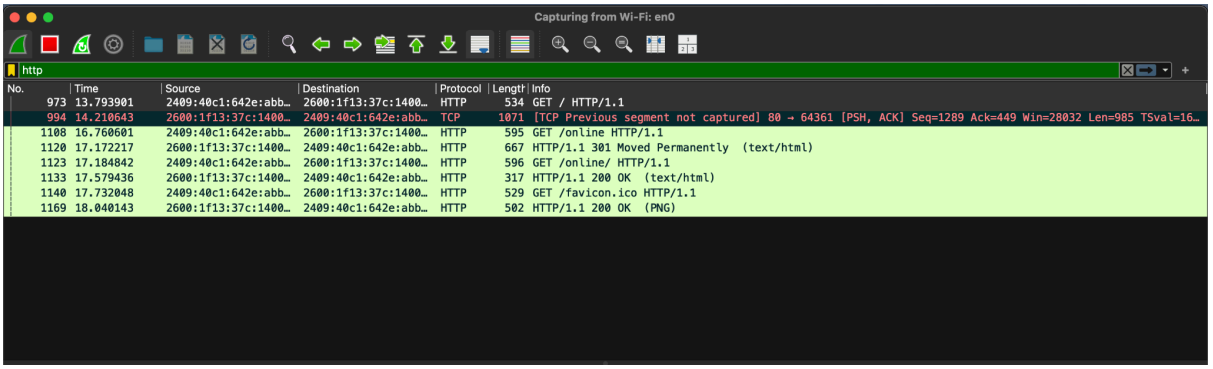
```
http
```

Procedure

1. Start capture.
2. Open an HTTP website.
3. Stop capture.
4. Apply HTTP filter.
5. Observe GET and Response packets.

Result

Successfully analyzed HTTP traffic.



Practical No. 6: HTTPS Analysis

Objective

To observe encrypted HTTPS communication.

Display Filter

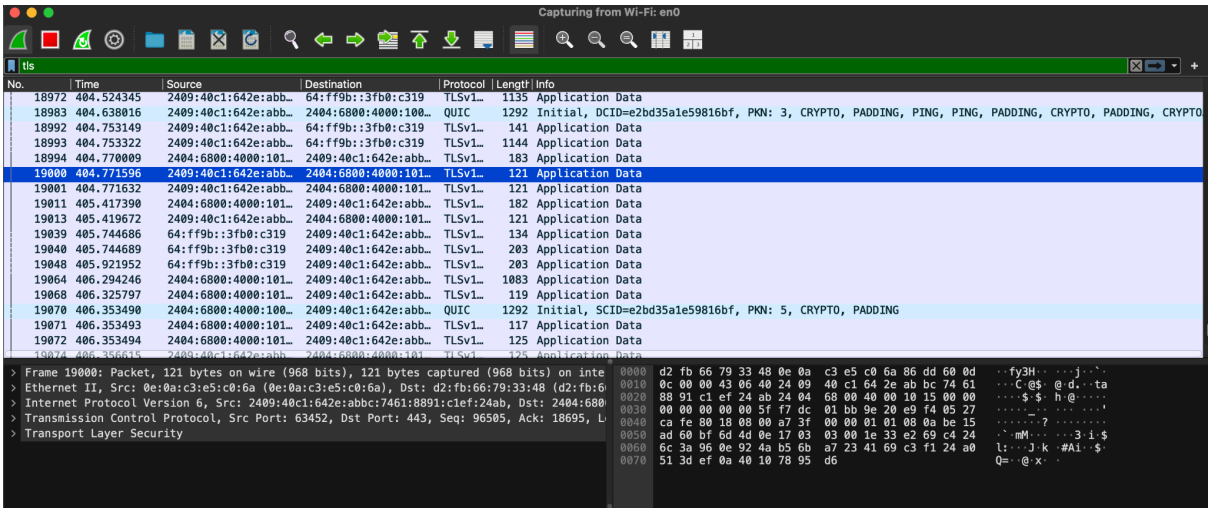
```
tls
```

Observation

Application data is encrypted and cannot be viewed without decryption keys.

Result

Successfully identified HTTPS traffic.



Practical No. 7: ICMP (Ping) Analysis

Objective

To analyze ICMP packets.

Display Filter

```
icmp
```

Procedure

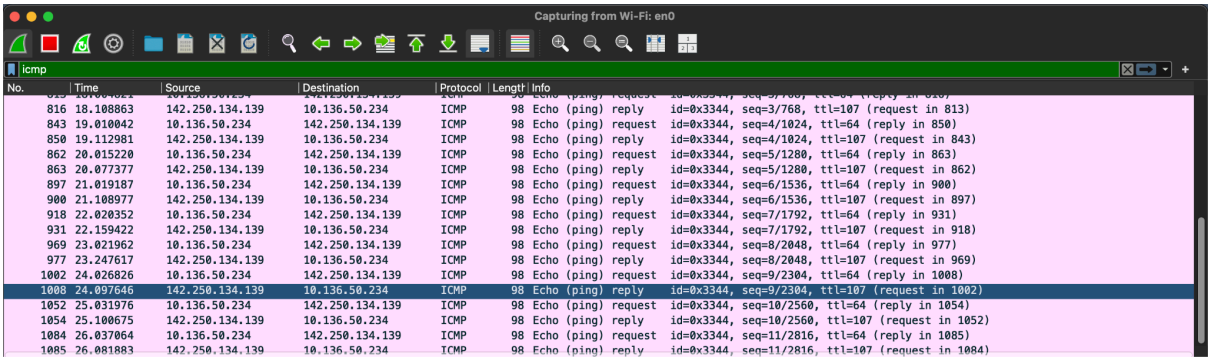
1. Start Wireshark.
2. Open Command Prompt.
3. Run:

```
ping google.com
```

4. Stop capture.
5. Observe Echo Request and Echo Reply packets.

Result

Successfully analyzed ICMP packets.



Practical No. 8: ARP Analysis

Objective

To analyze Address Resolution Protocol.

Display Filter

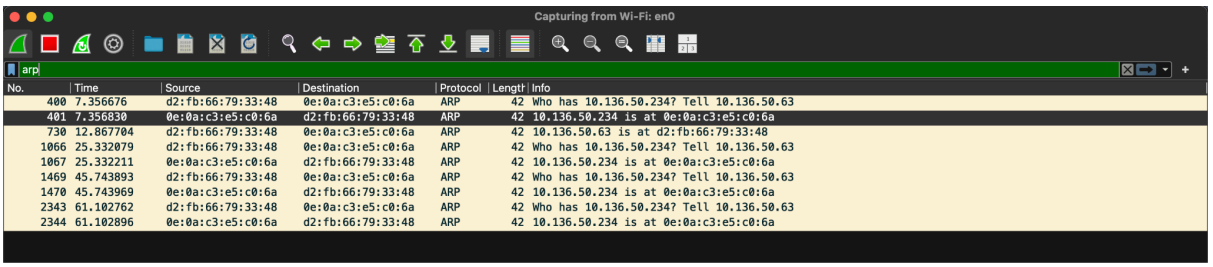
```
arp
```

Observation

ARP resolves IP addresses to MAC addresses.

Result

Successfully analyzed ARP packets.



Practical No. 9: DHCP Analysis

Objective

To observe DHCP communication.

Display Filter

`dhcp`

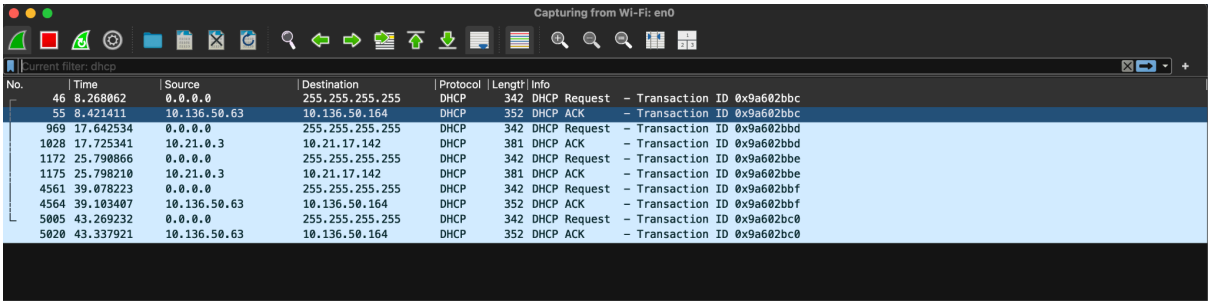
Observation

DHCP follows the DORA process:

- Discover
- Offer
- Request
- Acknowledge

Result

Successfully analyzed DHCP packets.



Practical No. 10: FTP Packet Analysis

Objective

To analyze FTP communication.

Display Filter

```
ftp
```

Observation

FTP transmits usernames and passwords in plaintext.

Result

Successfully analyzed FTP packets.

Useful Wireshark Filters

Protocol	Filter
TCP	`tcp`
UDP	`udp`
HTTP	`http`
HTTPS/TLS	`tls`
DNS	`dns`
ARP	`arp`
ICMP	`icmp`
DHCP	`dhcp`
FTP	`ftp`
SMTP	`smtp`

Common Viva Questions

1. What is Wireshark?

A network protocol analyzer used to capture and inspect packets.

2. What is packet capture?

The process of recording network traffic.

3. What is a display filter?

A filter that shows only packets matching specified criteria.

4. Difference between capture filter and display filter?

- Capture filter: Filters packets before capture.
- Display filter: Filters packets after capture.

5. What is TCP?

A reliable, connection-oriented transport protocol.

6. What is DNS?

A protocol that translates domain names into IP addresses.

7. What is ARP?

A protocol that maps IP addresses to MAC addresses.

8. What is ICMP?

A protocol used for diagnostics and error reporting, such as `ping`.

9. What is DHCP?

A protocol that automatically assigns IP addresses to devices.

10. What is the default Wireshark file format?

`.pcapng`

Conclusion

This practical manual demonstrated how to install Wireshark, capture live network traffic, analyze common network protocols (TCP, UDP, DNS, HTTP, HTTPS, ICMP, ARP, DHCP, and FTP), apply display filters, and interpret packet-level communication for network troubleshooting and protocol analysis.

If you want, I can also prepare this as a **fully formatted `.docx` file** with a cover page, table of contents, page numbers, and college-style formatting.